STAFF REVIEW · H446 217 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Index 2 in [1,3,6,8,7].

2. To stop on empty or single-element partitions.

3. O(n log n).

4. Left 0..2, right 4..6; do not include pivot index 3 again.

I Do — source worked steps

1. Predict the next state — Partition around 3, moving 1 to the left and placing pivot 3 at index 1.

2. Reveal and explain — Left segment [1] is already sorted; sort only the right segment [7,5].

3. Reveal and explain — The right recursion produces [5,7]; final array [1,3,5,7].

We Do — guided CHECK

1. Reveal and explain
The left [1,2] is already sorted; the right [6,5] needs another partition.
After sorting the right part, final array is [1,2,4,5,6].

You Do — source mark schemes

1. [3] Prevents calls on empty [1] or single-element ranges [1], stopping recursion correctly [1].

2. [6] Pivot=2 [1]; move 1 left [1] then place pivot to form [1,2,6,4] [2]; recursive right segment [6,4] sorts to [4,6] [1]; final [1,2,4,6] [1].

3. [6] Average O(n log n) [1] with balanced partitions [1]; worst $O(n^2)$ [1] with repeated unbalanced partitions [1]; in-place partitioning typically uses less additional array space than merge sort [1], though recursive quicksort still uses call-stack space [1].

Exit Ticket — source answer

1. False. The pivot is already fixed and is excluded from both calls.